

Francesco GAFFI

CURRENT MAIN POSITION: Assistant Professor of Statistics
Department of Economics
University of Bergamo, Italy
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ACADEMIC POSITIONS

APR 2026 - NOW	Research and Teaching Fellow , Bocconi Institute for Data Science and Analytics and Department of Decision Sciences, Bocconi University, Milano, Italy
NOV 2025 - NOW	Assistant Professor of Statistics (RTT STAT01/A), Department of Economics, University of Bergamo, Italy
AUG 2025 - OCT 2025	Researcher , Bocconi Institute for Data Science and Analytics, Milano, Italy
FEB 2024 - JUL 2025	Postdoctoral Associate , Department of Mathematics, University of Maryland, College Park (MD), USA
FEB 2023 - JAN 2024	Robert and Sara Lumpkins Postdoctoral Fellow , Department of Applied and Computational Mathematics and Statistics, University of Notre Dame, South Bend (IN), USA
FEB 2023 - NOW	External affiliate , BayesLab, Bocconi Institute for Data Science and Analytics, Milano, Italy
SEP 2022 - JAN 2023	Ph.D. Candidate , Department of Decision Sciences, Bocconi University, Milano, Italy
SEP 2018 - AUG 2022	Ph.D. Student in Statistics , Department of Decision Sciences, Bocconi University, Milano, Italy

INDUSTRY EXPERIENCE

2016 - 2017 | **Quantitative analyst intern**, Ladbroke's Coral, Roma, Italy

EDUCATION

2023	Ph.D. in STATISTICS, <i>cum laude</i> Bocconi University , Milano, Italy Supervisors: Antonio Lijoi and Igor Prünster Thesis: Modeling with discrete random probability measures Defence date: January 23rd, 2023
2018	M.Sc. in PURE AND APPLIED MATHEMATICS, <i>cum laude</i> Università di Roma Tor Vergata , Italy Supervisor: Domenico Marinucci Thesis: Functional data analysis on $\mathcal{L}^2(\mathbb{S}^2)$ Defence date: July 24th, 2018
2016	B.Sc. in MATHEMATICS Università di Roma Tor Vergata , Italy
2014	M.Mus. in ORCHESTRAL REPERTOIRE AND CHAMBER MUSIC, <i>cum laude</i> Conservatorio Licinio Refice , Frosinone, Italy
2011	B.Mus. in CLARINET Conservatorio Licinio Refice , Frosinone, Italy

Summer schools

JUL 2019 | *Random graphs and complex networks: structure and function*,
Bocconi Summer School in Advanced Statistics and Probability, Como, Italy

FUNDING

2026	Seedcorn Grant, structured member of interdepartmental research project: ‘ <i>ALABICCO: Accertamenti di LABORatorio Io e Criteri interpretativi per applicazioni forensi</i> ’, University of Bergamo
2024	Scientific Exchange support, Swiss National Science Foundation
2023	Robert and Sara Lumpkins Postdoctoral Fellowship (3.5 years), University of Notre Dame
2022	Special financial support for PhD candidates, Fondazione Invernizzi
2018	PhD Scholarship (4 years), Bocconi University

AWARDS

2026	ISBA2026 Travel Award, International Society for Bayesian Analysis
2025	BNP14 Travel Award, International Society for Bayesian Analysis
2024	ISBA2024 Travel Award, University of California, Los Angeles
2023	Student Paper Award, American Statistical Association - Section on Bayesian Statistical Science
2022	★ ISBA2022 Travel Award, International Society for Bayesian Analysis ★ BNP2022 Travel Award, International Society for Bayesian Analysis
2014	Mancini Award for Best Graduate Student - Winds Section, Conservatorio Licinio Refice

PUBLICATIONS

Preprints

GAFFI, F., JOSEPHS, N., LIN, L. **Exchangeable random permutations with an application to Bayesian graph matching.**
<https://arxiv.org/abs/2602.01993>

Refereed papers

FRANZOLINI, B., GAFFI, F. (2026). **Complexity bounds for Dirichlet process slice samplers.** *Proceedings of the 43rd International Conference on Machine Learning*, Proceedings of Machine Learning Research 306:31553-31574.
<https://proceedings.mlr.press/v306/franzolini26a.html>

DURANTE, D., GAFFI, F., LIJOI, A., PRÜNSTER, I. (2025). **Partially exchangeable stochastic block models for (node-colored) multilayer networks.** *Journal of the American Statistical Association*, 120(551), 1811–1827.
<https://doi.org/10.1080/01621459.2025.2507825>
[2023 ASA-SBSS Student Paper Award winner]

GAFFI, F., LIJOI, A., PRÜNSTER, I. (2025). **Random probability measures with fixed mean distributions.** *Annals of Applied Probability*, 35(4), 2239-2261.
<https://doi.org/10.1214/24-AAP2130>

Refereed short papers, book chapters and discussions

GAFFI, F., LEGRAMANTI, S., RIMELLA, L. (2026). **Gaffi, Legramanti, and Rimella’s contribution to the discussion of ‘Regression by Composition’ by Farewell et al.** *Journal of the Royal Statistical Society Series B: Statistical Methodology*.
<https://doi.org/10.1093/jrsssb/qkag089>

GAFFI, F., FRANZOLINI, B. (2026). **Contributed discussion on “Robust probabilistic inference via a constrained transport metric” by A. Chakraborty, A. Bhattacharya, and D. Pati.** *Bayesian Analysis*, 21(2), 1006–1008.
<https://doi.org/10.1214/25-BA1535>

GAFFI, F. (2026). **Conditionally partially exchangeable partitions for dynamic networks.** In *New Trends in Bayesian Statistics. BaYSM 2023*, 79-92, Springer Proceedings in Mathematics and Statistics, vol 511, Springer.

https://doi.org/10.1007/978-3-031-99009-0_7

CATALANO, M., GAFFI, F. (2025). **Contributed discussion on “A tree perspective on stick-breaking models in covariate-dependent mixtures”** by Horiguchi, A., Chan, C., and Ma, L. *Bayesian Analysis*, 20(3) 1201-1203.

<https://doi.org/10.1214/24-BA1462>

GAFFI, F. (2025). **Modeling multiple node-colored networks with partial exchangeability.** In *Statistics for Innovation III. SIS 2025*, 274-280, Italian Statistical Society Series on Advances in Statistics, Springer.

https://doi.org/10.1007/978-3-031-95995-0_46

GAFFI, F., LIJOI, A., PRÜNSTER, I. (2022). **Specification of the base measure of non-parametric priors via random means.** In *New Frontiers in Bayesian Statistics. BaYSM 2021*, 91-100, Springer Proceedings in Mathematics and Statistics, vol 405, Springer.

https://doi.org/10.1007/978-3-031-16427-9_9

Theses

GAFFI, F. (2023). **Modelling with discrete random probability measures.** *Ph.D. Thesis.* Bocconi University, Milano.

GAFFI, F. (2018). **Functional data analysis on $\mathcal{L}^2(\mathbb{S}^2)$.** *M.Sc. Thesis.* Università di Roma Tor Vergata, Roma.

Online repositories

- * R code for **Complexity bounds for Dirichlet process slice samplers**
<https://github.com/beatricefranzolini/DPalg>
- * R code for **Exchangeable random permutations with an application to Bayesian graph matching**
<https://github.com/francescogaffi/BayesGM>
- * R code for **Partially exchangeable stochastic block models for (node-colored) multilayer networks**
<https://github.com/francescogaffi/pexsbm>

EDITORIAL ACTIVITY

Reviewer	<i>Bayesian Analysis, Electronic Journal of Statistics, Journal of Complex Networks, Journal of Econometrics, Journal of Machine Learning Research, Journal of the American Statistical Association, Journal of the Royal Statistical Society Series B: Statistical Methodology, Journal of the Royal Statistical Society Series C: Applied Statistics, Statistics and Computing, Statistics and Probability Letters, Stochastic Processes and their Applications.</i>
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Committee member 2024-2026 ASA-SBSS Student Paper Award

VISITING PERIODS

OCT 2026	Department of Statistics, North Carolina State University, Raleigh (NC), USA
OCT 2026	Department of Mathematics, University of Maryland, College Park (MD), USA
OCT 2025	Brin Mathematics Research Center, University of Maryland, College Park (MD), USA

PRESENTATIONS

Seminars

OCT 2026	Statistics seminar, Department of Mathematics, University of Maryland, College Park, USA.
OCT 2026	Bayesian statistics seminar, Department of Statistics, North Carolina State University, USA.
MAR 2026	Brown bag seminar, Department of Economics, University of Bergamo, Italy.

Invited talks

JAN 2027	STING2027 , Statistical Inference on Networks and Graphs, Imperial College London, UK.
JUL 2026	BNP2026 , 4th Bayesian Nonparametrics Networking Workshop, University of Seoul, South Korea.
JUN 2026	ISNPS2026 , 7th International Symposium on Nonparametric Statistics, Thessaloniki, Greece. IMPMS2026 , 5th Italian Meeting on Probability and Mathematical Statistics, University of Palermo, Italy.
MAY 2026	StaTalk2026 , University of Torino, Italy.
JAN 2026	Nonparametric Bayesian Inference - Computational Issues , Institute for Computational and Experimental Research in Mathematics, Brown University, Providence (RI), USA.
DEC 2025	Bayesian Uncertainty Quantification in Large Models , Chennai Mathematical Institute, India.
SEP 2025	MSP2025 , The Mathematics of Subjective Probability, Università di Milano-Bicocca, Italy.
JUN 2025	BNP14 , 14th International Conference on Bayesian Nonparametrics, University of California Los Angeles, USA.
JAN 2025	Recent Developments in the Techniques of the Bayesian Paradigm , Banaras Hindu University, Varanasi, India.
SEP 2024	Frontiers of Bayesian Inference and Data Science , Casa Matemática Oaxaca, Mexico.
AUG 2024	Bernoulli-IMS2024 , 11th World Congress in Probability and Statistics, Ruhr-Universität Bochum, Germany.
JUL 2024	ISBA2024 , World Meeting of the International Society for Bayesian Analysis, Università Ca' Foscari, Venezia, Italy.
JUN 2024	Satellite workshop to ISBA World Meeting , Università della Svizzera Italiana, Lugano, Switzerland. IMPMS2024 , 4th Italian Meeting on Probability and Mathematical Statistics Roma, Italy.
AUG 2023	JSM2023 , Joint Statistical Meetings, Toronto, Canada. EcoSta2023 , 6th International Conference on Econometrics and Statistics, Waseda University, Tokyo, Japan.
JUN 2023	10th International Purdue Symposium on Statistics , Purdue University, West Lafayette (IN), USA.
DEC 2022	IISA2022 , International Indian Statistical Association Conference, Indian Institute of Science, Bengaluru, India.
JUN 2022	IMPMS2022 , 3rd Italian Meeting on Probability and Mathematical Statistics, Alma Mater Studiorum, Bologna, Italy.

JUN 2021	EcoSta2021 , 4th International Conference on Econometrics and Statistics, virtual.
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Contributed talks

JUN 2025	SIS2025 , Scientific Meeting of the Italian Statistical Society, Università di Genova, Italy.
MAY 2025	BISP14 , 14th Bayesian Inference for Stochastic Processes Workshop, Consiglio Nazionale delle Ricerche, Milano, Italy.
APR 2025	BaYSM2025 , Bayesian Young Statisticians Meeting, virtual.
NOV 2023	BaYSM2023 , Bayesian Young Statisticians Meeting, virtual.
JUN 2022	ISBA2022 , World Meeting of the International Society for Bayesian Analysis, Montréal, Canada.
APR 2022	BNP2022 , 1st Bayesian Nonparametrics Networking Workshop, University of Cyprus, Nicosia, Cyprus.
SEP 2021	BaYSM2021 , Bayesian Young Statisticians Meeting, virtual.
JUN 2021	ISBA2021 , World Meeting of the International Society for Bayesian Analysis, virtual. ISBA@CIRM , Junior session of ISBA2021 mirror event, Centre International de Rencontres Mathématiques, Marseille, France.

Poster sessions

JUL 2026	ICML2026 , International Conference on Machine Learning, Seoul, South Korea.
JUN 2026	ISBA2026 , World Meeting of the International Society for Bayesian Analysis, Nagoya, Japan.
JUN 2024	BaYSM2024 , Bayesian Young Statisticians Meeting, Università Ca' Foscari, Venezia, Italy.
JUN 2023	Approximation Methods in Bayesian Analysis , Centre International de Rencontres Mathématiques, Marseille, France.
OCT 2022	BNP13 , 13th International Conference on Bayesian Nonparametrics, Puerto Varas, Chile.

CONFERENCE ORGANIZATION

Scientific committee member

APR 2027	BaYSM2027 , Bayesian Young Statisticians Meeting, fully online.
JUN 2026	BaYSM2026 , Bayesian Young Statisticians Meeting, Chiba University, Japan.

Organizing committee member

APR 2027	BaYSM2027 , Bayesian Young Statisticians Meeting, fully online.
JUN 2027	BNP15 , 15th International Conference on Bayesian Nonparametrics, University of Bergamo, Italy (LOC).
JUN 2026	BaYSM2026 , Bayesian Young Statisticians Meeting, Chiba University, Japan.

Session organizer

MAY 2027	BayesComp2027 , Biennial meeting of the Bayesian Computation Section of the International Society for Bayesian Analysis, Texas A&M University, College Station (TX), USA.
DEC 2026	IISA2026 , International Indian Statistical Association Conference, Banaras Hindu University, Varanasi, India.
JUN 2026	IMPMS2026 , 5th Italian Meeting on Probability and Mathematical Statistics, University of Palermo, Italy.

SERVICE TO SCIENTIFIC SOCIETIES

Member	ISBA, International Society for Bayesian Analysis j-ISBA, Junior Section of ISBA BNP-ISBA, Bayesian Nonparametrics Section of ISBA IMS, Institute of Mathematical Statistics SIS, Società Italiana di Statistica
Chair-Elect	j-ISBA (2026)
Section Chair	j-ISBA (2027)

SERVICE TO THE DEPARTMENT

2026-NOW	Departmental coordinator for English Additional Learning Requirements
2026-NOW	Degree programme coordinator for Mathematics Additional Learning Requirements

RESEARCH SUPERVISION

2026	David Verdelli, <i>M.Sc. thesis: 'Youth market trajectories in Italy (2017-2021): a linked survey-administrative data approach'</i> , M.Sc. in Economics and Data Analysis, University of Bergamo.
2025	Edoardo Stefanelli, Research Assistant, Bocconi Institute for Data Science and Analytics, Bocconi University (<i>with Beatrice Franzolini</i>).

TEACHING

Experience

2026 - NOW	Lecturer , Bocconi University Bayesian statistical methods, M.Sc. in Economics (1 semester)
2026 - NOW	Lecturer , University of Bergamo ★ Statistical inference, B.Sc. in Statistics (2 semesters) ★ Statistics, B.Eng. in Management Engineering (1 semester) ★ Machine Learning for Finance, M.Sc. in Finance (1 semester)
2024 - 2025	Lecturer , University of Maryland, College Park ★ Applied probability and statistics, B.Sc. in Computer Science (3 semesters) ★ High-dimensional statistical models, Ph.D. in Statistics (2 guest lectures)
2023 - 2024	Lecturer , University of Notre Dame ★ Probability, B.Sc. in Applied Mathematics and Statistics (1 semester)
2020 - 2023	Adjunct lecturer , Bocconi University ★ Probability, B.Sc. in Mathematics for AI (2 semesters) ★ Probability, Ph.D. in Economics and Finance (3 semesters) ★ Statistics, B.Sc. in Economics (2 semesters)
2019 - 2023	Teaching assistant , Bocconi University ★ Optimal control, M.Sc. in Data Science (3 semesters) ★ Statistics, B.Sc. in Economics (4 semesters) ★ Stochastic processes, M.Sc. in Data Science (4 semesters) ★ Probability, B.Sc. in Mathematics for AI (2 semesters)
2017 - 2018	Teaching assistant , Università di Roma Tor Vergata ★ Discrete mathematics, B.Sc. in Computer Science

Publications

FORTINI, S., GAFFI, F. (2021). **Introduction to Probability. BAI 30456.** BAI series. Volume 2. Egea, Milano.

Certifications

2019	Bocconi Excellence in Advanced Teaching https://bestr.it/award/show/8b3a93e2027f4cbb8988f46cd3ba6371a0e5d075
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